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Fluorescence lifetime imaging in a flame

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Abstract

A novel method for two-dimensional fluorescence lifetime imaging is presented. The technique is demonstrated on averaged planar laser-induced fluorescence (PLIF) signals recorded in a flame. Although demonstrated on averaged images, the concept applies equally well to single-shot images. Formaldehyde was probed through pico-second excitation at 355 nm in a rich methane/oxygen flame ($\Phi = 2.6$). Images were recorded with a dual ICCD camera detection setup with different gate characteristics of the two cameras. The recorded images were analyzed using simulated values of LIF signal detection to generate two-dimensional images of effective lifetimes. Measured lifetimes range from roughly 1 to 4.5 ns. The lifetime image data were used for quenching correction of the LIF images, which, after correction, showed better qualitative agreement compared to a formaldehyde concentration profile simulated with the GRI 3.0 mechanism.
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1. Introduction

Laser-induced fluorescence (LIF) is one of the most widely used optical measurement techniques in combustion diagnostics. Among its strengths are imaging possibility, high sensitivity, high spatial resolution and that it provides species specificity in many measurement situations. For an overview of LIF in combustion studies, see the review article by Kose-Höinghaus [1]. Despite the fact that emitted fluorescence intensity holds information about the concentration of the probed species, LIF is generally used as a qualitative measurement tool. For many applications, such as monitoring where certain reactions take place or observing the dynamic behavior of specific quantum states, qualitative measurements are enough.

The extension to quantitative measurements is not trivial. Most parameters that influence the relation between measured signal and absolute concentration can be calibrated. Disregarding saturation, the problem generally arises with the determination of the fluorescence quantum yield, which is affected by collision processes, stimulated emission, predissociation, photoionization and chemical reactions involving the excited molecules. In combustion related studies, uncertainties regarding fluorescence quantum yield are almost exclusively due to varying collision environments. LIF techniques aiming at measurements free from collisional quenching include saturated LIF, laser-induced predissociative fluorescence and time-gated short pulse LIF [2,3]. The first two techniques aim at making stimulated emission and predissociation, respectively, the dominant loss mechanisms, thus allowing for collisional

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quenching to be neglected. The idea behind the short-pulse technique is to record only the initial part of the fluorescence decay curve, i.e. to measure on a timescale which is short enough to prevent collision processes to largely affect the recorded signal. However, to fully compensate for differences in fluorescence quantum yield, the temporal evolution of the fluorescence decay ought to be recorded.

An obvious way of obtaining an image of the fluorescence lifetime is to record the fluorescence at different delay times after excitation, as done by Dowling et al. and Bormann et al. [4,5]. They did measurements on dyes as well as on endogenous fluorophores in collagen and elastin, their outlooks being within the biomedical field. Ossler et al. [6] used a streak camera and recorded 2D-images in streak mode as well as in focus mode. Through a mathematical relationship they were able to obtain images of fluorescence lifetimes down to about 0.9 ns, and they estimated a lifetime resolution of 200–250 ps for their setup. The work by Ossler et al. is a proof-of-concept study, demonstrating lifetime measurements of plain copy paper dyed with a green-line marker. Ni and Melton [7] showed proof-of principle 1D measurements of lifetimes ranging from roughly 6 to 50 ns. Later they published two follow-up articles on 2D-lifetime imaging [8,9]. The shortest measured lifetime was around 14 ns.

For combustion studies, the imaging technique ought to be applicable in non-steady flow situations where the lifetimes are around or shorter than 10 ns. This request puts requirements on short laser pulses and at least one detector with a fast gate, accompanied by a rigorous evaluation routine. In this paper, we present a novel method for two-dimensional fluorescence lifetime imaging. The technique is demonstrated on planar laser-induced fluorescence (PLIF) signals recorded in a laminar flame. Formaldehyde (CH_2O) was probed using excitation by the third harmonic (355 nm) of a picosecond Nd:YAG laser. Although demonstrated on averaged images, the measurement concept applies equally well to single-shot images.

2. Experimental

The experimental setup, which is seen in Fig. 1, consisted of an Nd:YAG laser (Ekspla PL 2143C), sheet forming laser optics, a 70/30 quartz beam splitter (BS) and two ICCD cameras (Princeton Instruments, PI Max II). The third harmonic (355 nm) from the laser was used and the pulses, with Gaussian shaped temporal profiles, were 30 ps (FWHM) long. The laser profile was expanded by about a factor of two while passing through a telescope (ST), consisting of two spherical lenses, before being focused by a cylindrical lens

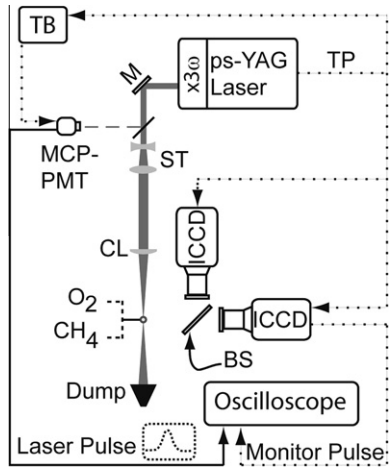

Fig. 1. A schematic image of the experimental setup. The flame is represented by a circle in the focus of the laser sheet.

(CL) with 400 mm focal length, forming a ~20 mm high vertical laser sheet. The laser-pulse energy was about 15 mJ in the probe volume. One of the cameras had a short-pulse gate option, allowing the gate FWHM to be less than 2 ns. The other camera gate was set to 50 ns. Nikkor 50 mm 1:1.2 lenses were used as collection optics and 3 mm thick GG385 Schott filters where used to reduce laser stray light. Triggering of the camera gates was performed through a pre-triggering signal sent out by the laser ~170 ns ahead of the actual laser pulse. The time position of the laser pulse was measured with a gated PMT (MCP-PMT; Hamamatsu R5916U-50, rise time ~170 ps) connected to a 3 GHz digital oscilloscope (Lecroy Wavemaster 8300). The monitor pulses of the camera gates were also displayed on the oscilloscope, and hence the time jitter between the laser pulse and the camera gates could be observed and measured. The time jitter was found to be 135 ps (RMS).

When performing spectral investigations, the beam splitter was removed along with the camera objective of the ICCD camera located orthogonally to the laser beam direction. A spectrometer (Acton Research Corporation) with a 1200 groves/mm grating was placed in front of this ICCD camera. A slit width of 200 μm was chosen and a 3 mm thick GG385 Schott filter was used for stray light suppression.

The burner consisted of a 2 mm diameter gas-welding nozzle and the flame was operated on a premixed gas flow of methane and oxygen with a fuel equivalence ratio of 2.6. Due to the stability of the inner flame cone, averaging was not considered to reduce the spatial resolution to any considerable extent. Hence, series of 50 images with 90 single shots accumulated (on chip) in each recorded image were acquired in order to improve the signal-to-noise ratio.

3. Measurement concept

The aim of this study was to extract effective lifetimes from 2D-PLIF measurements. The basic idea of the technique is to evaluate experimental results using calculated values for detection of simulated LIF signals and thereby extract lifetimes of the experimental PLIF images. In the experiments, LIF signals were detected using two ICCD cameras with different gate characteristics. One camera was operated with a long gate, while the other camera was operated with a short gate. The gate of the short-gated camera was positioned for the gate to open before the laser pulse reached the probe volume, thus only the closing flank of that gate overlapped the LIF signal in time. The gate of the camera with long gate was opened before the beginning of the LIF signal and remained opened for a time period exceeding the time window during which the LIF signal could be measured.

Grid images were recorded with the two cameras in the same positions as during flame measurements. In order to overlap images recorded with the two cameras, an image-transformation was performed using an in-house Matlab code based on optimization by simulated annealing [10]. The overlap algorithm contains two different parts. The first part is a loop that tries to find an optimized transformation based on image rotation, stretch, translation and correction for barrel/pincushion distortion. The second part is a loop that performs a higher order non-linear transformation on the image to improve the overlap beyond what is possible with the first loop. The transform function obtained from the grid images was then applied to all images recorded with the short-gated camera. Equation (1), below, was then used to form an image $R(x, y)$, which is a ratio involving the image recorded with the short gate, $I_S(x, y)$, and the image recorded with the long gate, $I_L(x, y)$:

$$R(x,y) = I_s(x,y)/[I_s(x,y) + I_L(x,y)]. \quad (1)$$

By forming the ratio given in Eq. (1), inhomogeneities in the laser profile and concentration variations are cancelled, thus minimizing errors due to signal variations stemming from these effects.

To be able to calculate the detected signal of both cameras, the gate functions need to be known. Therefore, the gate functions of the two cameras were extracted from images of Rayleigh scattering recorded with sequentially shifting gate-delay time. The time step between two consecutive measurement points was 1 ns for the long gate and 0.1 ns for the short gate. Each measurement point was based on 25 accumulated single shots. Apart from showing the actual shapes of the gates, these functions also show the relative transmissions of the two camera gates. In order to compensate for differences in spectral sensitivity between the Rayleigh wavelength and the mean fluorescence wavelength of the two cameras, images obtained from shifting the gate-delay time while monitoring the LIF signal were recorded as well. Now, by comparing the LIF intensity with the Rayleigh intensity at the gate center, the differences in wavelength sensitivity between the two cameras was obtained.

The simulated LIF signals, $S(t, \tau)$, used for determination of the lifetimes of the measured LIF signals, were convolutions of a Gaussian function, representing the laser pulse, and single-exponential decay functions. The decay constant is denoted τ. By integrating the product of $S(t, \tau)$ and the camera gate function, $G_i(t, d)$ in time, a simulation of the measured signal is given by the following expression:

$$I_i(d,\tau) = \int S(t,\tau) G_i(t,d)\, dt. \quad (2)$$

The camera gate delay is assigned d, whereas the subscript i is an index referring to either the short or the long gate function. A graphical description of the simulated signal detection is seen in Fig. 2. The modeled LIF signals, $S(t, \tau)$, with $\tau = 10$ ns, are displayed as dashed curves. Temporal descriptions of the measured gate functions are shown as solid black curves, in Fig. 2a for the short-gated camera, and in Fig. 2b for the long-gated camera. The gray areas represent the integrated signals; $I_S(d, \tau)$ and $I_L(d, \tau)$, using the short and long camera gate, respectively. With $I_S(d, \tau)$ and $I_L(d, \tau)$ inserted into Eq. (1), signal ratios for a set of τ values can be determined. The set of τ values should cover a range wider than the range of expected lifetimes. The relation between the signal ratio and τ is expressed by a function $\tau(R)$. The set of LIF lifetimes used in the present simulation spans from 0.5 to 10 ns with a time step of 0.1 ns. Finally, combining Eq. (1) and $\tau(R)$ for each pixel, a lifetime image is obtained.

4. Results and discussion

A spectral investigation was performed to study the spectral components of the laser-induced signal. Due to the limited resolution of the flame structure using the current experimental configuration, the spatial origin of the different spectral components was indistinguishable. Spontaneous Raman scattering, both from methane and oxygen are seen in the spectrum in Fig. 3. However, these components contribute only marginally to the over-all signal, especially since the predominant part of the Raman signal from methane is obtained in the cold, unburned fuel region and hence only partly overlaps spatially with the formaldehyde signal. The other peaks were, after comparison with the literature [11–13], found to constitute a formaldehyde LIF spectrum. It

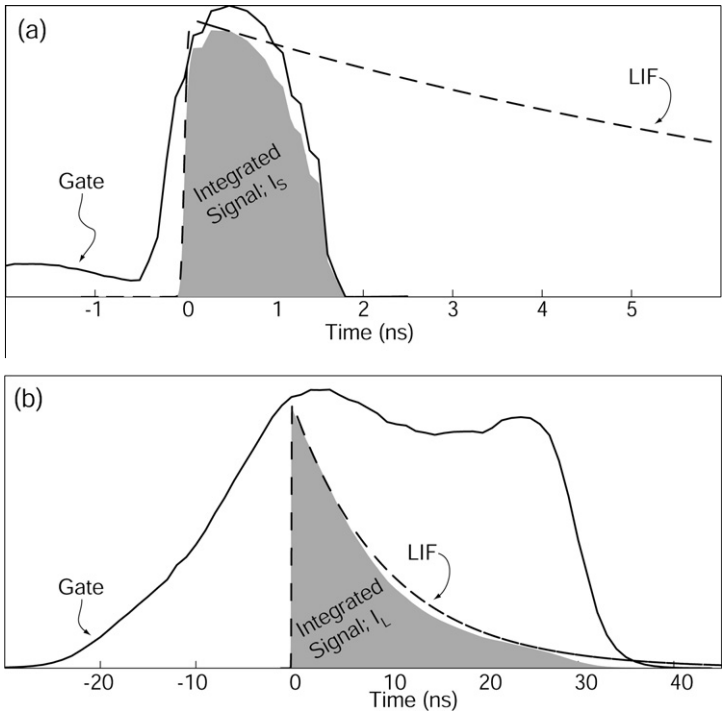

Fig. 2. Graphical illustration of the simulated LIF signal for a 2 ns camera gate width in (a) and a 50 ns camera gate width in (b). The dashed curves are the simulated LIF signal with a 10 ns decay constant. The solid black curves display the experimentally mapped gate functions that were used in the simulations. The shaded gray areas are the total integrated LIF signals from the simulation, calculated by Eq. (2).

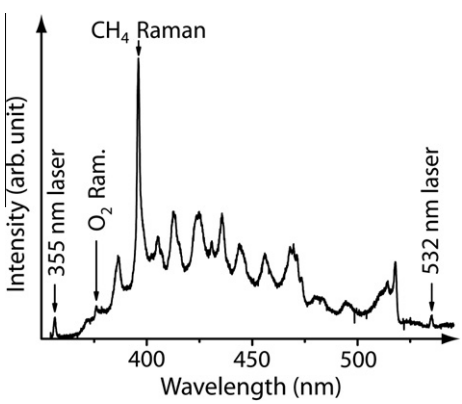

Fig. 3. The spectral representation of the experimental signal clearly shows a formaldehyde LIF spectrum. The underlying bump is probably due to a combination of non-resolved formaldehyde peaks and interfering fluorescence from PAH molecules. Raman emission peaks from methane and oxygen are labeled in the spectrum.

should be pointed out that the LIF spectrum of polycyclic aromatic hydrocarbons (PAH), most often formed in rich hydrocarbon flames, overlaps that of formaldehyde [14]. This interference, in combination with Doppler line broadening of the formaldehyde spectrum and limited spectral resolution could explain the background contribution seen in the spectrum.

The gate functions measured and used in the simulations are seen in Fig. 2a (short gate, G_s) and Fig. 2b (long gate, G_L). It should be pointed out that the gates shown in Fig. 2 are not the actual gate functions but convolutions between the temporal profile of the laser pulse, the probability density function of the time jitter, and the averaged gate functions of the cameras. For a thorough review of the theory, see the article by Ehn et al. [15]. The ratio between the differences in spectral sensitivity for the Rayleigh and the LIF wavelengths of the two cameras, including collection optics was found to be ~2.5. This number implies that a spectral sensitivity correction of the gate functions is necessary for the present cameras in order to obtain correct LIF detection simulations.

The lifetime of the LIF signal, τ, as a function of the ratio, R (Eq. (1)), for the simulated LIF signals, is seen in Fig. 4. When applying the simulated results to the experimental data to find the LIF lifetime for the ratio in each pixel, it is decisive that τ has a determinate solution to every R. The problem of ambiguous solutions arises if simulations are performed for short lifetimes

(hundreds of picoseconds) and the laser excitation occurs before the short camera gate function has reached its maximum value, which is the case in Fig. 2a. Due to this ambiguity, simulations of LIF signals with lifetimes shorter than 300 ps could not be determined. Such short lifetimes are, however, believed to be below the current measurement limit.

An averaged PLIF image (4500 accumulations) recorded by the camera with the long gate, $I_L(x, y)$, corrected for the fluence variations within the laser sheet profile is seen in Fig. 5a. By inserting the images recorded with the two cameras into Eq. (1), the image $R(x, y)$ was generated. Each pixel value of the image obtained from Eq. (1) was evaluated using $\tau(R)$ shown in Fig. 4. The result is an image of the effective lifetimes of the signal, shown in Fig. 5b. As seen in this figure, the effective lifetimes of the LIF signal span from about 1 ns to roughly 4.5 ns. The lifetime image and the LIF data were used to reproduce a temporal description of the total LIF signal. This simulated signal was an exponential decay curve with an effective lifetime of 3.3 ns. An MCP-PMT was used to verify the lifetime of the total LIF signal, detecting an exponential curve with the effective lifetime of 3.4 ns. Fluorescence lifetime measurements of formaldehyde in the preheat zone of a premixed methane/air flame was carried out by Tobai and Dreier [16]. They presented fluorescence lifetimes spanning from 7 to 18 ns. The shorter fluorescence lifetimes measured in this study are due to the fact that we used pure oxygen instead of air, which changes the quenching conditions substantially [17]. A contributing factor to shorter lifetimes could also be the interfering Raman signal from methane, which basically has the shape of the laser pulse (~30 ps). This interference could bias the evaluation of the fluorescence lifetimes, leading to shorter lifetime predictions.

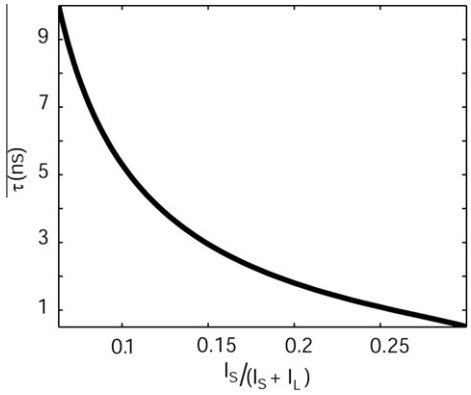

Fig. 4. Calculated relations between the LIF lifetime (τ) and the ratio (R) presented in Eq. (1), which is used for every pixel value of $R(x, y)$ to generate the lifetime of the LIF signal.

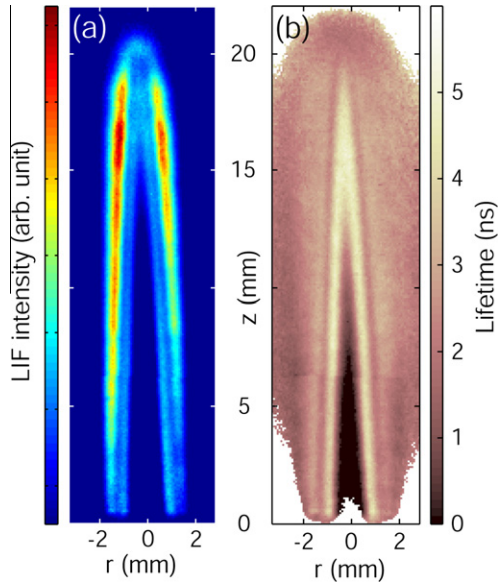

Fig. 5. (a) Accumulated PLIF image (4500 shots) of formaldehyde in the flame. (b) Effective lifetime image of the LIF signal in (a).

However, the Raman signal is very weak and only dominates the inner part of the cold unburned fuel region. In this region the lifetime-evaluation model collapses since the Raman signal is merely 30 ps and has the shape of a Gaussian. A spectral filter could be used to suppress the Raman signal, e.g. a Schott glass GG435. Such a filter was not chosen since it also would reduce the formaldehyde signal, which is spectrally overlapping with the Raman interference from methane (Fig. 3).

Single-shot evaluation was undertaken. However, due to the weak absorption cross-section of formaldehyde at the excitation wavelength, software binning, 2 in x-direction and 10 in y-direction, was performed. A set of 50 single shot lifetime images was analyzed. Standard deviation of the fluorescence lifetime was calculated to about 1.5 ns, in the parts of the PLIF image where the formaldehyde LIF signal was pronounced. Better spectral overlap between the laser and the molecular transition, using a tunable pico-second laser system would result in higher signal-to-noise ratio, and open up the possibility to probe other molecules, e.g. NO, OH and CH.

When studying the two images shown in Fig. 5, there seems to be a double structure in the LIF signal, parallel to the reaction zone. The fluorescence lifetimes of these two peaks differ. The inner structure has a longer lifetime, and the intensity will therefore be overpredicted if it is not corrected for differences in fluorescence lifetimes. To correct the PLIF image, $I_L(x, y)$, shown in Fig. 5a, for intensity differences due to variations in the fluorescence lifetime, each pixel was divided by

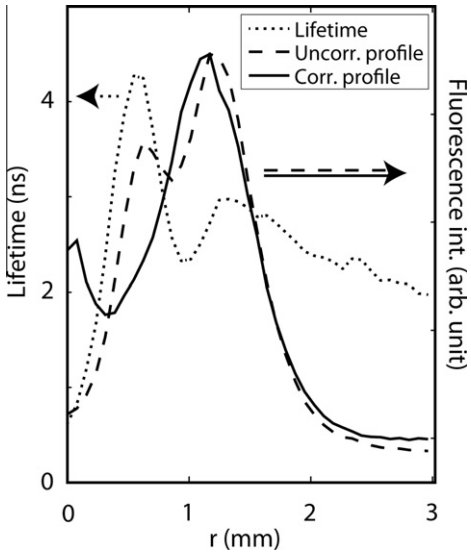

Fig. 6. Profile curves of the flame at height 8 mm in Fig. 5. The formaldehyde LIF (uncorrected and corrected) are normalized with their peak values.

$I_L(d, \tau)$, determined from Eq. (2) with τ-values taken from the lifetime image.

In Fig. 6, the average of 10 pixel rows at a flame height of 8 mm is shown for the LIF signal lifetime, corrected and uncorrected LIF signals. A flame temperature profile was extracted from Rayleigh scattering measurements. Maximum temperature was found to be roughly 2200 K. By studying the LIF lifetime, represented by the dotted curve in Fig. 6, the maximum lifetime is found in a region where the temperature is below 1000 K, whereas the maximum formaldehyde LIF signal is found in a somewhat hotter region. After correcting the LIF signal for differences in fluorescence lifetimes, only one peak, instead of two, is seen in the measured relative concentration profile. This result indicates that the inner peak structure, seen in Fig. 5a, is an effect of longer lifetimes of the LIF signal in the colder region, and that the concentration profile should have merely one peak. A concentration profile based on flame-speed calculations for a premixed laminar flame using the GRI 3.0 mechanism [18], supports this result, showing one single concentration peak of formaldehyde.

To conduct a quantitative LIF experiment, a calibration measurement is needed to account for system parameters. Further correction of the LIF signal by accounting for population distribution differences due to strong temperature variations [19] would improve the accuracy of the concentration measurement.

It should be mentioned that the corrected formaldehyde LIF profile, shown in Fig. 6, shows higher values than the uncorrected LIF signal in the unburned central region of the flame. This ghost signal is due to the interfering Raman scattering from methane, which strongly enhances the corrected LIF signal due to the inherent short lifetime of the Raman scattering process. In general, short lived interfering laser-induced signals, such as Rayleigh or Raman scattering, results in an underprediction of the lifetimes, whereas a continuos background, such as chemiluminescence would lead to overpredictions of the fluorescence lifetime.

5. Conclusion

A conceptually novel method for two-dimensional fluorescence lifetime imaging has been presented. The technique is successfully demonstrated on planar laser-induced fluorescence (PLIF) of formaldehyde recorded in a premixed, laminar methane/oxygen flame. Measured lifetimes range from about 1 to 4.5 ns. An MCP-PMT measurement verified these values. The results show that concentration profiles of combustion species measured with LIF may be rather misleading if not compensated for differences in fluorescence lifetimes. After correcting for variations in fluorescence lifetimes, the results obtained in the current study qualitatively agree fairly well with a formaldehyde concentration profile, simulated with the GRI 3.0 mechanism.

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